

Anexo balances de masa y energía detallados Proyecto Análisis

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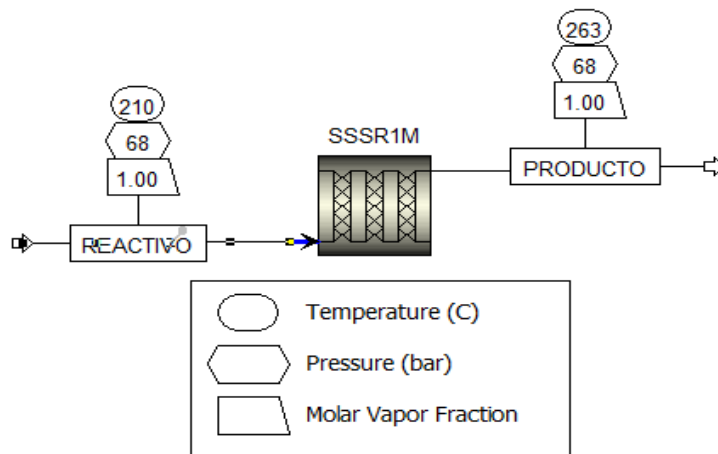
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Balance en el Reactor:

Los balances de masa y energía un poco mas detallados se realizaron en el Software Aspen plus; a continuación, se muestra el reporte filtrado y la totalidad de datos arrojados por el simulador, así como el dimensionamiento del volumen y peso de la carcasa para su posterior uso en el calculo del CAPEX:



Cinética utilizada:

$$r_{\text{CH}_3\text{OH}} = \frac{k_1 P_{\text{CO}_2} P_{\text{H}_2} - k_6 P_{\text{H}_2\text{O}} P_{\text{CH}_3\text{OH}} P_{\text{H}_2}^{-2}}{(1 + k_2 P_{\text{H}_2\text{O}} P_{\text{H}_2}^{-1} + k_3 P_{\text{H}_2}^{0.5} + k_4 P_{\text{H}_2\text{O}})^3} \quad \left[\frac{\text{kmol}}{\text{kg}_{\text{cat}} \text{s}} \right] \quad (\text{A.1})$$

Reverse water gas shift reaction

$$r_{\text{RWGS}} = \frac{k_5 P_{\text{CO}_2} - k_7 P_{\text{H}_2\text{O}} P_{\text{CO}} P_{\text{H}_2}^{-1}}{1 + k_2 P_{\text{H}_2\text{O}} P_{\text{H}_2}^{-1} + k_3 P_{\text{H}_2}^{0.5} + k_4 P_{\text{H}_2\text{O}}} \quad \left[\frac{\text{kmol}}{\text{kg}_{\text{cat}} \text{s}} \right] \quad (\text{A.2})$$

$$\ln k_i = A_i + \frac{B_i}{T} \quad (\text{A.3})$$

Table A.1
Parameters of the rearranged kinetic model.

k_1	A_1	-29.87
	B_1	4811.2
k_2	A_2	8.147
	B_2	0
k_3	A_3	-6.452
	B_3	2068.4
k_4	A_4	-34.95
	B_4	14,928.9
k_5	A_5	4.804
	B_5	-11,797.5
k_6	A_6	17.55
	B_6	-2249.8
k_7	A_7	0.1310
	B_7	-7023.5

Se introdujeron los parámetros cinéticos así:

1) CO₂ + 3 H₂ --> CH₃OH(MIXED) + H₂O(MIXED)

Reacting phase: Vapor Rate basis: Cat (wt)

LHHW kinetic expression

$$r = \frac{[\text{Kinetic factor}][\text{Driving force expression}]}{[\text{Adsorption expression}]}$$

Kinetic factor

If To is specified Kinetic factor = $k(T/T_o)^n e^{-(E/R)[1/T-1/T_o]}$

If To is not specified Kinetic factor = $kT^n e^{-E/RT}$

k: 1

n: 0

E: 0 cal/mol

To: C

Solids

Driving Force

Adsorption

Driving Force Expression

Reacting phase: Vapor

[C] basis: Partial pressure

Enter term: Term 1

Term 1

Concentration exponents for reactants		Concentration exponents for products	
Component	Exponent	Component	Exponent
CO ₂	1	CH ₃ OH	0
H ₂	1	H ₂ O	0

Coefficients for driving force constant

A: -29.87 B: 4811.2 C: 0 D: 0

Close

Driving Force Expression

Reacting phase: Vapor

[C] basis: Partial pressure

Enter term: Term 2

Term 2

Concentration exponents for reactants		Concentration exponents for products	
Component	Exponent	Component	Exponent
CO ₂	0	CH ₃ OH	1
H ₂	-2	H ₂ O	1

Coefficients for driving force constant

A: 17.55 B: -2249.8 C: 0 D: 0

Close

Adsorption Expression

Reacting phase: Vapor

[C] basis: Partial pressure

Adsorption expression exponent: 3

Component	Term No. 1	Term No. 2	Term No. 3	Term No. 4	Term No. 5
H ₂	0	-1	0.5	0	
H ₂ O	0	1	0	1	

Term no.	1	2	3	4
Coefficient A	0	8.147	-6.452	-34.95
Coefficient B	0	0	2068.4	14928.9
Coefficient C	0	0	0	0
Coefficient D	0	0	0	0

Close

De manera análoga se introdujeron los datos para la reacción de RWGSm, obteniendo los siguientes resultados (resultados ya organizados)

Datos filtrados:

Material			
Stream Name	Units	PRODUCTO	REACTIVO
Mole Flows	kmol/hr	1208.127724	1323
H ₂	kmol/hr	788.7713409	991
CO ₂	kmol/hr	242.6436165	330
CH ₃ OH	kmol/hr	57.43613783	0
CO	kmol/hr	31.92024566	2

H2O	kmol/hr	87.35638349	0
Mole Fractions			
H2		0.652887377	0.749055178
CO2		0.200842685	0.249433107
CH3OH		0.047541445	0
CO		0.026421251	0.001511716
H2O		0.072307242	0

Reactor dimensions length [meter]	7
Reactor dimensions diameter [meter]	1.5

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
REACTOR	45800	204000	8600	22138

Datos Crudos que arroja Aspen:

A continuación, se presentan TODOS los datos arrojados por el simulador los datos esenciales para el balance de masa y energía se encuentran en la sección superior; sin embargo se decide añadir esta sección por si se desea conocer algún otro dato en específico.

CORRIENTES			
Material			
Stream Name	Units	PRODUCTO	REACTIVO
Description			
From		SSSR1M	
To			SSSR1M
Stream Class		CONVEN	CONVEN
Maximum Relative Error			
Cost Flow	\$/hr		
MIXED Substream			
Phase		Vapor Phase	Vapor Phase
Temperature	C	263.1439224	210
Pressure	bar	67.99999988	68
Molar Vapor Fraction		1	1
Molar Liquid Fraction		0	0
Molar Solid Fraction		0	0
Mass Vapor Fraction		1	1
Mass Liquid Fraction		0	0
Mass Solid Fraction		0	0
Molar Enthalpy	cal/mol	-24190.16723	-22089.80475
Mass Enthalpy	cal/gm	-1762.974362	-1762.974362
Molar Entropy	cal/mol-K	-3.314612996	-3.40586794
Mass Entropy	cal/gm-K	-0.241568307	-0.271820323
Molar Density	mol/cc	0.001505345	0.001662192

Mass Density	gm/cc	0.020655173	0.020827012
Enthalpy Flow	cal/sec	-8118003.247	-8118003.247
Average MW		13.72122462	12.52985025
Mole Flows	kmol/hr	1208.127724	1323
H2	kmol/hr	788.7713409	991
CO2	kmol/hr	242.6436165	330
CH3OH	kmol/hr	57.43613783	0
CO	kmol/hr	31.92024566	2
H2O	kmol/hr	87.35638349	0
Mole Fractions			
H2		0.652887377	0.749055178
CO2		0.200842685	0.249433107
CH3OH		0.047541445	0
CO		0.026421251	0.001511716
H2O		0.072307242	0
Mass Flows	kg/hr	16576.99188	16576.99188
H2	kg/hr	1590.068371	1997.73708
CO2	kg/hr	10678.69703	14523.234
CH3OH	kg/hr	1840.377918	0
CO	kg/hr	894.0988489	56.0208
H2O	kg/hr	1573.749708	0
Mass Fractions			
H2		0.0959202	0.120512642
CO2		0.644187867	0.876107927
CH3OH		0.111020017	0
CO		0.053936134	0.003379431
H2O		0.094935783	0
Volume Flow	l/min	13375.98054	13265.61868
Vapor Phase			
Molar Enthalpy	cal/mol	-24190.16723	-22089.80475
Mass Enthalpy	cal/gm	-1762.974362	-1762.974362
Molar Entropy	cal/mol-K	-3.314612996	-3.40586794
Mass Entropy	cal/gm-K	-0.241568307	-0.271820323
Molar Density	mol/cc	0.001505345	0.001662192
Mass Density	gm/cc	0.020655173	0.020827012
Enthalpy Flow	cal/sec	-8118003.247	-8118003.247
Average MW		13.72122462	12.52985025
Mole Flows	kmol/hr	1208.127724	1323
H2	kmol/hr	788.7713409	991
CO2	kmol/hr	242.6436165	330
CH3OH	kmol/hr	57.43613783	0
CO	kmol/hr	31.92024566	2
H2O	kmol/hr	87.35638349	0
Mole Fractions			
H2		0.652887377	0.749055178
CO2		0.200842685	0.249433107
CH3OH		0.047541445	0

MEA	kmol/hr	0.16	232.78	0.00	0.00	232.78	162.95	69.84	232.95	162.95	70.00
CH4	kmol/hr	443.07	59.76	461.00	0.00	59.76	41.83	17.93	41.84	41.83	0.00
Mole Fractions											
H2		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2		0.0385	0.4972	0.3997	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA		0.0004	0.4001	0.0000	0.0000	0.7957	0.7957	0.7957	0.8478	0.7957	1.0000
CH4		0.9611	0.1027	0.6003	0.0000	0.2043	0.2043	0.2043	0.1522	0.2043	0.0000

Dimensionamiento: para su posterior uso en el CAPEX

Equipment				
Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]
DESTABS	27500	209300	3500	19872
ABS	32800	186400	4100	15735

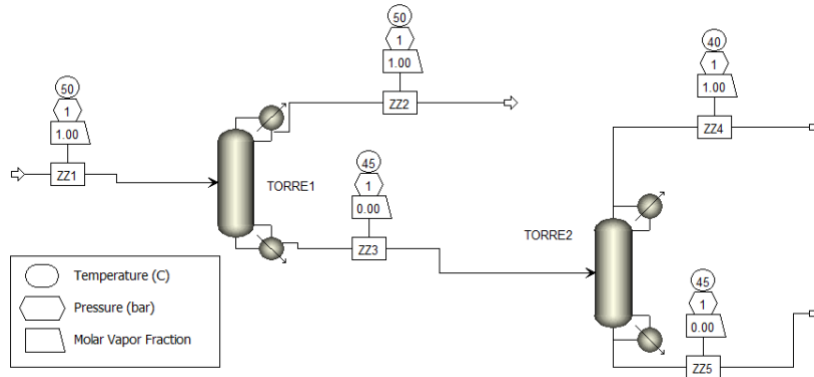
Datos Crudos que arroja Aspen:

Material											
Stream Name	Units	AB1	AB2	BIOGAS	DT1	DT2	DV1	DV2	FEED	HT1	MEA
Description											
From		ABS	ABS		DESTABS	DESTABS	SPLITTER	SPLITTER	MIXER	HEXCH	
To			DESTABS	ABS		SPLITTER	HEXCH		ABS	MIXER	MIXER
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Maximum Relative Error											
Cost Flow	\$ /hr										
MIXED Substream											
Phase		Vapor Phase	Liquid Phase	Vapor Phase	Liquid Phase	Liquid Phase	Liquid Phase	Liquid Phase			Liquid Phase
Temperature	C	25.3441	-85.7364	30.0000	-88.1486	-88.1486	-88.1841	-88.1841	25.4709	25.0000	25.0000
Pressure	bar	1.5000	2.0000	2.0000	1.2000	1.2000	2.0000	2.0000	2.0000	2.0000	2.0000
Molar Vapor Fraction		1.0000	0.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.1402	0.1928	0.0000
Molar Liquid Fraction		0.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	0.8598	0.8072	1.0000
Molar Solid Fraction		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Mass Vapor Fraction		1.0000	0.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0415	0.0597	0.0000
Mass Liquid Fraction		0.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	0.9585	0.9403	1.0000
Mass Solid Fraction		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Molar Enthalp	cal/mol	-20749.2512	-81392.7763	-48223.8016	-100980.9096	-59204.3113	-59204.3113	-59204.3113	-57197.4642	-54794.4144	-64228.4249
Mass Enthalp	cal/gm	-1210.8267	-1696.7938	-1771.4820	-2294.5096	-1141.1218	-1141.1218	-1141.1218	-1054.7921	-1056.1242	-1051.4819
Molar Entrop	cal/mol-K	-18.9602	-80.1494	-11.1797	-34.5978	-116.6586	-116.6652	-116.6652	-102.9221	-97.9199	-117.5865
Mass Entrop	cal/gm-K	-1.1064	-1.6709	-0.4107	-0.7861	-2.2485	-2.2486	-2.2486	-1.8980	-1.8873	-1.9250
Molar Density	mol/cc	0.0001	0.0204	0.0001	0.0277	0.0156	0.0156	0.0156	0.0006	0.0004	0.0132
Mass Density	kg/cum	1.0383	979.0907	2.1720	1218.8581	811.0760	811.0964	811.0964	30.1399	21.2349	808.8982
Enthalpy Flow	cal/sec	-2657056.8829	-13153636.9599	-10287744.3371	-8113178.9821	-4811134.2931	-3367794.0051	-1443340.2879	-4365834.1219	-3116940.2447	-1248886.0391
Average MW		17.1364	47.9686	27.2223	44.0098	51.8826	51.8826	51.8826	54.2263	51.8826	61.0837
Mole Flows	kmol/hr	461.0000	581.7850	768.0000	289.2373	292.5477	204.7834	87.7643	274.7850	204.7834	70.0000
H2	kmol/hr	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2	kmol/hr	17.7627	289.2373	307.0000	289.2373	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH	kmol/hr	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO	kmol/hr	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O	kmol/hr	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA	kmol/hr	0.1646	232.7847	0.0000	0.0000	232.7847	162.9493	69.8354	232.9493	162.9493	70.0000
CH4	kmol/hr	443.0727	59.7630	461.0000	0.0000	59.7630	41.8341	17.9289	41.8357	41.8341	0.0000
Mole Fractions											
H2		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2		0.0385	0.4972	0.3997	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA		0.0004	0.4001	0.0000	0.0000	0.7957	0.7957	0.7957	0.8478	0.7957	1.0000
CH4		0.9611	0.1027	0.6003	0.0000	0.2043	0.2043	0.2043	0.1522	0.2043	0.0000
Mass Flows	kg/sec	2.1944	7.7521	5.8074	3.5359	4.2161	2.9513	1.2648	4.1390	2.9513	1.1877
H2	kg/sec	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2	kg/sec	0.2171	3.5359	3.7531	3.5359	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH	kg/sec	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO	kg/sec	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O	kg/sec	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA	kg/sec	0.0028	3.9498	0.0000	0.0000	3.9498	2.7649	1.1849	3.9526	2.7649	1.1877
CH4	kg/sec	1.9745	0.2663	2.0544	0.0000	0.2663	0.1864	0.0799	0.1864	0.1864	0.0000
Mass Fractions											
H2		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2		0.0990	0.4561	0.6463	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA		0.0013	0.5095	0.0000	0.0000	0.9368	0.9368	0.9368	0.9550	0.9368	1.0000
CH4		0.8998	0.0344	0.3537	0.0000	0.0632	0.0632	0.0632	0.0450	0.0632	0.0000
Volume Flow	l/min	126809.9588	475.0563	160429.4344	174.0601	311.8927	218.3194	93.5655	8239.6827	8338.9955	88.1005
Vapor Phase											
Molar Enthalp	cal/mol	-20749.2512		-48223.8016					-17811.5847	-17815.2889	
Mass Enthalp	cal/gm	-1210.8267		-1771.4820					-1109.4054	-1109.6691	
Molar Entrop	cal/mol-K	-18.9602		-11.1797					-20.6210	-20.6340	
Mass Entrop	cal/gm-K	-1.1064		-0.4107					-1.2844	-1.2852	
Molar Density	mol/cc	0.0001		0.0001					0.0001	0.0001	
Mass Density	gm/cc	0.0010		0.0022					0.0013	0.0013	
Enthalpy Flow	cal/sec	-2657056.8829		-10287744.3371					-190540.6966	-195405.2103	
Average MW		17.1364		27.2223					16.0551	16.0546	
Mole Flows	kmol/hr	461.0000		768.0000					38.5113	39.4862	
H2	kmol/hr	0.0000		0.0000					0.0000	0.0000	
CO2	kmol/hr	17.7627		307.0000					0.0000	0.0000	
CH3OH	kmol/hr	0.0000		0.0000					0.0000	0.0000	
CO	kmol/hr	0.0000		0.0000					0.0000	0.0000	
H2O	kmol/hr	0.0000		0.0000					0.0000	0.0000	
MEA	kmol/hr	0.1646		0.0000					0.0105	0.0104	
CH4	kmol/hr	443.0727		461.0000					38.5007	39.4759	
Mole Fractions											
H2		0.0000		0.0000					0.0000	0.0000	
CO2		0.0385		0.3997					0.0000	0.0000	
CH3OH		0.0000		0.0000					0.0000	0.0000	
CO		0.0000		0.0000					0.0000	0.0000	
H2O		0.0000		0.0000					0.0000	0.0000	
MEA		0.0004		0.0000					0.0003	0.0003	
CH4		0.9611		0.6003					0.9997	0.9997	
Mass Flows	kg/hr	7899.8958		20906.7210					618.3010	633.9356	
H2	kg/hr	0.0000		0.0000					0.0000	0.0000	
CO2	kg/hr	781.7334		13511.0086					0.0000	0.0000	
CH3OH	kg/hr	0.0000		0.0000					0.0000	0.0000	
CO	kg/hr	0.0000		0.0000					0.0000	0.0000	
H2O	kg/hr	0.0000		0.0000					0.0000	0.0000	
MEA	kg/hr	10.0533		0.0000					0.6431	0.6338	
CH4	kg/hr	7108.1090		7395.7124					617.6579	633.3018	
Mass Fractions											
H2		0.0000		0.0000					0.0000	0.0000	
CO2		0.0990		0.6463					0.0000	0.0000	
CH3OH		0.0000		0.0000					0.0000	0.0000	
CO		0.0000		0.0000					0.0000	0.0000	
H2O		0.0000		0.0000					0.0000	0.0000	
MEA		0.0013		0.0000					0.0010	0.0010	
CH4		0.8998		0.3537					0.9990	0.9990	
Volume Flow	l/min	126809.9588		160429.4344					7944.2695	8132.3744	
Liquid Phase											
Molar Enthalp	cal/mol		-81392.7763		-100980.9096	-59204.3113	-59204.3114	-59204.3114	-63617.4014	-63628.0015	-64228.4249
Mass Enthalp	cal/gm		-1696.7938		-2294.5096	-1141.1218	-1141.1218	-1141.1218	-1052.4277	-1052.7266	-1051.4819

Molar Entrop cal/mol-K		-80.1494		-34.5978	-116.6586	-116.6652	-116.6652	-116.3372	-116.3820	-117.5865
Mass Entropy cal/gm-K		-1.6709		-0.7861	-2.2485	-2.2486	-2.2486	-1.9246	-1.9255	-1.9250
Molar Density mol/cc		0.0204		0.0277	0.0156	0.0156	0.0156	0.0133	0.0133	0.0132
Mass Density gm/cc		0.9791		1.2189	0.8111	0.8111	0.8111	0.8057	0.8059	0.8089
Enthalpy Flow cal/sec		-13153636.9599		-8113178.9821	-4811134.2931	-3367794.0123	-1443340.2910	-4175310.8229	-2921535.0344	-1248886.0391
Average MW		47.9686		44.0098	51.8826	51.8826	51.8826	60.4482	60.4411	61.0837
Mole Flows	kmol/hr	581.7850		289.2373	292.5477	204.7834	87.7643	236.2737	165.2971	70.0000
H2	kmol/hr	0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2	kmol/hr	289.2373		289.2373	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH	kmol/hr	0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO	kmol/hr	0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O	kmol/hr	0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA	kmol/hr	232.7847		0.0000	232.7847	162.9493	69.8354	232.9401	162.9389	70.0000
CH4	kmol/hr	59.7630		0.0000	59.7630	41.8341	17.9289	3.3336	2.3582	0.0000
Mole Fractions										
H2		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2		0.4972		1.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA		0.4001		0.0000	0.7957	0.7957	0.7957	0.9859	0.9857	1.0000
CH4		0.1027		0.0000	0.2043	0.2043	0.2043	0.0141	0.0143	0.0000
Mass Flows	kg/hr	27907.3945		12729.2752	15178.1193	10624.6835	4553.4358	14282.3290	9990.7479	4275.8604
H2	kg/hr	0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2	kg/hr	12729.2752		12729.2752	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH	kg/hr	0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO	kg/hr	0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH4	kg/hr	958.7627		0.0000	958.7627	671.1339	287.6288	53.4798	37.8321	0.0000
Mass Fractions										
H2		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO2		0.4561		1.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH3OH		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CO		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H2O		0.0000		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA		0.5095		0.0000	0.9368	0.9368	0.9368	0.9963	0.9962	1.0000
CH4		0.0344		0.0000	0.0632	0.0632	0.0632	0.0037	0.0038	0.0000
Volume Flow	l/min	475.0563		174.0601	311.8927	218.3194	93.5655	295.4467	206.6211	88.1005

Balance en el tren de separación a la salida del reactor:

A continuación, se presentan los balances de masa y energía del tren de separación de productos a la salida del Reactor:



Datos filtrados:

Material								
Stream Name	Units	PRODUCTO	REACTIVO	ZZ1	ZZ2	ZZ3	ZZ4	ZZ5
Mole Flows	kmol/hr	1208.13	1323.00	1208.13	1063.48	144.65	56.89	87.76
H2	kmol/hr	788.77	991.00	788.77	788.77	0.00	0.00	0.00
CO2	kmol/hr	242.64	330.00	242.64	242.64	0.00	0.00	0.00
CH3OH	kmol/hr	57.44	0.00	57.44	0.06	57.38	56.80	0.57
CO	kmol/hr	31.92	2.00	31.92	31.92	0.00	0.00	0.00
H2O	kmol/hr	87.36	0.00	87.36	0.09	87.27	0.09	87.18
Mole Fractions								
H2		0.653	0.749	0.653	0.742	0.000	0.000	0.000
CO2		0.201	0.249	0.201	0.228	0.000	0.000	0.000
CH3OH		0.048	0.000	0.048	0.000	0.397	0.998	0.007
CO		0.026	0.002	0.026	0.030	0.000	0.000	0.000
H2O		0.072	0.000	0.072	0.000	0.603	0.002	0.993

Dimensionamiento:

Equipment						
Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]	Sizing Errors	Evaluation Errors
TORRE1	37400	205300	5300	19574		
TORRE2	22100	139100	2600	10646		

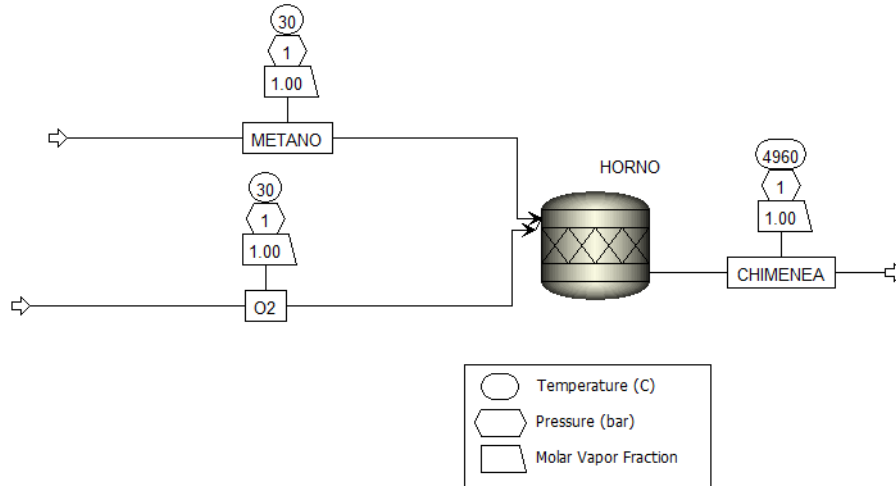
Datos Crudos que arroja Aspen:

Material								
Stream Name	Units	PRODUCTO	REACTIVO	ZZ1	ZZ2	ZZ3	ZZ4	ZZ5
Description								
From		SSSR1M			TORRE1	TORRE1	TORRE2	TORRE2
To			SSSR1M	TORRE1		TORRE2		
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Maximum Relative Error								
Cost Flow	\$/hr							
MIXED Substream								
Phase		Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Liquid Phase	Vapor Phase	Liquid Phase
Temperature	C	263.1439224	210	50	50	45	40.00587914	45.00587914
Pressure	bar	67.99999988	68	1.2	1.2	1.2	1	1
Molar Vapor Fraction		1	1	1	1	0	1	0
Molar Liquid Fraction		0	0	0	0	1	0	1
Molar Solid Fraction		0	0	0	0	0	0	0
Mass Vapor Fraction		1	1	1	1	0	1	0
Mass Liquid Fraction		0	0	0	0	1	0	1
Mass Solid Fraction		0	0	0	0	0	0	0
Molar Enthalpy	cal/mol	-24190.1672	-22089.8048	-25846.8053	-22061.0585	-63733.263	-47887.7709	-67864.673
Mass Enthalpy	cal/gm	-1762.97436	-1762.97436	-1883.7098	-1781.9382	-2702.9159	-1495.52805	-3747.98113
Molar Entropy	cal/mol-K	-3.314613	-3.40586794	0.776898994	2.370061639	-45.0825466	-30.4165587	-37.9692165
Mass Entropy	cal/gm-K	-0.24156831	-0.27182032	0.056620237	0.191437023	-1.91194246	-0.94990466	-2.09693645
Molar Density	mol/cc	0.001505345	0.001662192	4.47244E-05	4.46723E-05	0.027637744	3.92645E-05	0.040874186
Mass Density	gm/cc	0.020655173	0.020827012	0.000613673	0.00055306	0.651682719	0.001257276	0.740108648
Enthalpy Flow	cal/sec	-8118003.25	-8118003.25	-8673956.14	-6517081.76	-2560797.7	-756788.85	-1654305.94
Average MW		13.72122462	12.52985025	13.72122462	12.3803724	23.5794473	32.02064365	18.10699434
Mole Flows	kmol/hr	1208.127724	1323	1208.127724	1063.479996	144.6477288	56.8921837	87.75554509
H2	kmol/hr	788.7713409	991	788.7713409	788.7713409	0	0	0
CO2	kmol/hr	242.6436165	330	242.6436165	242.6436165	0	0	0
CH3OH	kmol/hr	57.43613783	0	57.43613783	0.057436138	57.37870169	56.80491468	0.573787017
CO	kmol/hr	31.92024566	2	31.92024566	31.92024566	0	0	0
H2O	kmol/hr	87.35638349	0	87.35638349	0.087356383	87.2690271	0.087269027	87.18175808
Mole Fractions								
H2		0.652887377	0.749055178	0.652887377	0.741688931	0	0	0
CO2		0.200842685	0.249433107	0.200842685	0.228160019	0	0	0
CH3OH		0.047541445	0	0.047541445	5.40077E-05	0.396678898	0.998466063	0.00653847
CO		0.026421251	0.001511716	0.026421251	0.0300149	0	0	0
H2O		0.072307242	0	0.072307242	8.2142E-05	0.603321102	0.001533937	0.99346153
Mass Flows	kg/hr	16576.99188	16576.99188	16576.99188	13166.27838	3410.713499	1821.724341	1588.989158
H2	kg/hr	1590.068371	1997.73708	1590.068371	1590.068371	0	0	0
CO2	kg/hr	10678.69703	14523.234	10678.69703	10678.69703	0	0	0
CH3OH	kg/hr	1840.377918	0	1840.377918	1.840377918	1838.53754	1820.152165	18.3853754
CO	kg/hr	894.0988489	56.0208	894.0988489	894.0988489	0	0	0
H2O	kg/hr	1573.749708	0	1573.749708	1.573749708	1572.175959	1.572175959	1570.603783
Mass Fractions								
H2		0.0959202	0.120512642	0.0959202	0.120768248	0	0	0
CO2		0.644187867	0.876107927	0.644187867	0.811064199	0	0	0
CH3OH		0.111020017	0	0.111020017	0.00013978	0.53904778	0.999136985	0.011570485
CO		0.053936134	0.003379431	0.053936134	0.067908244	0	0	0
H2O		0.094935783	0	0.094935783	0.000119529	0.46095222	0.000863015	0.988429515
Volume Flow	l/min	13375.98054	13265.61868	450212.2496	396770.6994	87.22837564	24149.09598	35.7827904
Vapor Phase								
Molar Enthalpy	cal/mol	-24190.1672	-22089.8048	-25846.8053	-22061.0585		-47887.7709	
Mass Enthalpy	cal/gm	-1762.97436	-1762.97436	-1883.7098	-1781.9382		-1495.52805	
Molar Entropy	cal/mol-K	-3.314613	-3.40586794	0.776898994	2.370061639		-30.4165587	
Mass Entropy	cal/gm-K	-0.24156831	-0.27182032	0.056620237	0.191437023		-0.94990466	
Molar Density	mol/cc	0.001505345	0.001662192	4.47244E-05	4.46723E-05		3.92645E-05	
Mass Density	gm/cc	0.020655173	0.020827012	0.000613673	0.00055306		0.001257276	
Enthalpy Flow	cal/sec	-8118003.25	-8118003.25	-8673956.14	-6517081.76		-756788.85	
Average MW		13.72122462	12.52985025	13.72122462	12.3803724		32.02064365	
Mole Flows	kmol/hr	1208.127724	1323	1208.127724	1063.479996		56.8921837	
H2	kmol/hr	788.7713409	991	788.7713409	788.7713409		0	

CO2	kmol/hr	242.6436165	330	242.6436165	242.6436165	0	
CH3OH	kmol/hr	57.43613783	0	57.43613783	0.057436138	56.80491468	
CO	kmol/hr	31.92024566	2	31.92024566	31.92024566	0	
H2O	kmol/hr	87.35638349	0	87.35638349	0.087356383	0.087269027	
Mole Fractions							
H2		0.652887377	0.749055178	0.652887377	0.741688931	0	
CO2		0.200842685	0.249433107	0.200842685	0.228160019	0	
CH3OH		0.047541445	0	0.047541445	5.40077E-05	0.998466063	
CO		0.026421251	0.001511716	0.026421251	0.0300149	0	
H2O		0.072307242	0	0.072307242	8.2142E-05	0.001533937	
Mass Flows	kg/hr	16576.99188	16576.99188	16576.99188	13166.27838	1821.724341	
H2	kg/hr	1590.068371	1997.73708	1590.068371	1590.068371	0	
CO2	kg/hr	10678.69703	14523.234	10678.69703	10678.69703	0	
CH3OH	kg/hr	1840.377918	0	1840.377918	1.840377918	1820.152165	
CO	kg/hr	894.0988489	56.0208	894.0988489	894.0988489	0	
H2O	kg/hr	1573.749708	0	1573.749708	1.573749708	1.572175959	
Mass Fractions							
H2		0.0959202	0.120512642	0.0959202	0.120768248	0	
CO2		0.644187867	0.876107927	0.644187867	0.811064199	0	
CH3OH		0.111020017	0	0.111020017	0.00013978	0.999136985	
CO		0.053936134	0.003379431	0.053936134	0.067908244	0	
H2O		0.094935783	0	0.094935783	0.000119529	0.000863015	
Volume Flow	l/min	13375.98054	13265.61868	450212.2496	396770.6994	24149.09598	
Liquid Phase							
Molar Enthalpy	cal/mol					-63733.263	-67864.673
Mass Enthalpy	cal/gm					-2702.9159	-3747.98113
Molar Entropy	cal/mol-K					-45.0825466	-37.9692165
Mass Entropy	cal/gm-K					-1.91194246	-2.09693645
Molar Density	mol/cc					0.027637744	0.040874186
Mass Density	gm/cc					0.651682719	0.740108648
Enthalpy Flow	cal/sec					-2560797.7	-1654305.94
Average MW						23.5794473	18.10699434
Mole Flows	kmol/hr					144.6477288	87.75554509
H2	kmol/hr					0	0
CO2	kmol/hr					0	0
CH3OH	kmol/hr					57.37870169	0.573787017
CO	kmol/hr					0	0
H2O	kmol/hr					87.2690271	87.18175808
Mole Fractions							
H2						0	0
CO2						0	0
CH3OH						0.396678898	0.00653847
CO						0	0
H2O						0.603321102	0.99346153
Mass Flows	kg/hr					3410.713499	1588.989158
H2	kg/hr					0	0
CO2	kg/hr					0	0
CH3OH	kg/hr					1838.53754	18.3853754
CO	kg/hr					0	0
H2O	kg/hr					1572.175959	1570.603783
Mass Fractions							
H2						0	0
CO2						0	0
CH3OH						0.53904778	0.011570485
CO						0	0
H2O						0.46095222	0.988429515
Volume Flow	l/min					87.22837564	35.7827904

Balance en Horno:

A continuación, se presentan los balances en el Horno, tener en cuenta que esta es una primera aproximación si bien la temperatura de salida da alrededor de los 4000 °C aún no se ha hecho integración energética para aprovecharla, de igual forma, se planea dividir dicho horno en mas sistemas para obtener múltiples corrientes, pero con temperaturas mas bajas.



Datos filtrados:

Material				
Stream Name	Units	CHIMENEA	METANO	O2
Mole Flows	kmol/hr	197.84	65.94	131.9
CARBO-01	kmol/hr	65.94	0	0
METHA-01	kmol/hr	0	65.94	0
OXYGE-01	kmol/hr	0.02	0	131.9
WATER	kmol/hr	131.88	0	0
Mole Fractions				
CARBO-01		0.33329964	0	0
METHA-01		0	1	0
OXYGE-01		0.00010109	0	1
WATER		0.66659927	0	0

Dimensionamiento

Equipment						
Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]	Sizing Errors	Evaluation Errors
HORNO	39900	217300	6400	18976		

Datos Crudos que arroja Aspen:

		Material		
Stream Name	Units	CHIMENE	METANO	O2
Description				
From		HORNO		
To			HORNO	HORNO
Stream Class		CONVEN	CONVEN	CONVEN
Maximum Relative Error				
Cost Flow	\$/hr			
MIXED Substream				
Phase		Vapor Phase	Vapor Phase	Vapor Phase
Temperature	C	4960.36073	30	30
Pressure	bar	1.01325	1.01325	1.01325
Molar Vapor Fraction		1	1	1
Molar Liquid Fraction		0	0	0
Molar Solid Fraction		0	0	0
Mass Vapor Fraction		1	1	1
Mass Liquid Fraction		0	0	0
Mass Solid Fraction		0	0	0
Molar Enthalpy	cal/mol	-5897.61461	-17760.3387	32.8480737
Mass Enthalpy	cal/gm	-221.044574	-1107.06255	1.0265408
Molar Entropy	cal/mol-K	27.4266338	-19.1180114	0.11106451
Mass Entropy	cal/gm-K	1.02795943	-1.19169092	0.0034709
Molar Density	mol/cc	2.3285E-06	4.0286E-05	4.0236E-05
Mass Density	gm/cc	6.2127E-05	0.0006463	0.0012875
Enthalpy Flow	cal/sec	-324106.688	-325310.204	1203.51692
Average MW		26.6806577	16.04276	31.9988
Mole Flows	kmol/hr	197.84	65.94	131.9
CARBO-01	kmol/hr	65.94	0	0
METHA-01	kmol/hr	0	65.94	0
OXYGE-01	kmol/hr	0.02	0	131.9
WATER	kmol/hr	131.88	0	0
Mole Fractions				
CARBO-01		0.33329964	0	0
METHA-01		0	1	0
OXYGE-01		0.00010109	0	1
WATER		0.66659927	0	0
WATER		0.45010032	0	0
Volume Flow	l/min	1416057.66	27279.8851	54636.2206
Vapor Phase				
Molar Enthalpy	cal/mol	-5897.61461	-17760.3387	32.8480737
Mass Enthalpy	cal/gm	-221.044574	-1107.06255	1.0265408
Molar Entropy	cal/mol-K	27.4266338	-19.1180114	0.11106451
Mass Entropy	cal/gm-K	1.02795943	-1.19169092	0.0034709
Molar Density	mol/cc	2.3285E-06	4.0286E-05	4.0236E-05
Mass Density	gm/cc	6.2127E-05	0.0006463	0.0012875
Enthalpy Flow	cal/sec	-324106.688	-325310.204	1203.51692
Average MW		26.6806577	16.04276	31.9988
Mole Flows	kmol/hr	197.84	65.94	131.9
CARBO-01	kmol/hr	65.94	0	0
METHA-01	kmol/hr	0	65.94	0
OXYGE-01	kmol/hr	0.02	0	131.9
WATER	kmol/hr	131.88	0	0
Mole Fractions				
CARBO-01		0.33329964	0	0
METHA-01		0	1	0
OXYGE-01		0.00010109	0	1
WATER		0.66659927	0	0
Mass Flows	kg/hr	5278.50131	1057.85959	4220.64172
CARBO-01	kg/hr	2902.00621	0	0
METHA-01	kg/hr	0	1057.85959	0
OXYGE-01	kg/hr	0.639976	0	4220.64172
WATER	kg/hr	2375.85513	0	0
Mass Fractions				
CARBO-01		0.54977844	0	0
METHA-01		0	1	0
OXYGE-01		0.00012124	0	1
WATER		0.45010032	0	0
Volume Flow	l/min	1416057.66	27279.8851	54636.2206